

Lab Exam:01

Title: Association/Classification/Fact and Dimension Table

Course title: Data Mining Problem Description

Course code: CSE-452
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Submitted to-

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1. Classification On my project dataset “Intelligence Automatic Supervisor Allocation System ”

The screenshot shows the Weka Explorer interface with the J48 classifier selected. The test options are set to cross-validation with 10 folds and a percentage split of 66%. The classifier output displays the J48 pruned tree structure, which is a simple decision tree based on Final_Marks. The summary statistics show a 100% accuracy rate, and the detailed accuracy by class table shows perfect classification for all four classes: Good, Poor, Average, and Excellent.

Classifier output

```
==== Run information ====
Scheme:   weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: ThesisHub_J48_500_Dataset
Instances: 500
Attributes: 6
  CGPA
  Attendance
  Submission_Days
  Teacher_Rating
  Final_Marks
  Performance_Class
Test mode: 10-fold cross-validation

==== Classifier model (full training set) ====

J48 pruned tree
-----
Final_Marks <= 84
| Final_Marks <= 74
| | Final_Marks <= 64: Poor (6.0)
| | Final_Marks > 64: Average (132.0)
| | Final_Marks > 74: Good (221.0)
| Final_Marks > 84: Excellent (141.0)

Number of Leaves :    4
Size of the tree :    7

Time taken to build model: 0.01 seconds

==== Stratified cross-validation ====
==== Summary ====

Correctly Classified Instances      500      100 %
Incorrectly Classified Instances    0        0 %
Kappa statistic                     1
Mean absolute error                 0
Root mean squared error             0
Relative absolute error             0 %
Root relative squared error         0 %
Total Number of Instances          500

==== Detailed Accuracy By Class ====

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Good
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Poor
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Average
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Excellent
Weighted Avg.   1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000

==== Confusion Matrix ====
```

Status: OK

This screenshot shows the same Weka Explorer interface as the first, but with the confusion matrix displayed at the bottom of the classifier output. The confusion matrix shows that all instances were correctly classified, with zero misclassifications across all classes.

Classifier output

```
==== Run information ====
Scheme:   weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: ThesisHub_J48_500_Dataset
Instances: 500
Attributes: 6
  CGPA
  Attendance
  Submission_Days
  Teacher_Rating
  Final_Marks
  Performance_Class
Test mode: 10-fold cross-validation

==== Classifier model (full training set) ====

J48 pruned tree
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Number of Leaves :    4
Size of the tree :    7

Time taken to build model: 0.01 seconds

==== Stratified cross-validation ====
==== Summary ====

Correctly Classified Instances      500      100 %
Incorrectly Classified Instances    0        0 %
Kappa statistic                     1
Mean absolute error                 0
Root mean squared error             0
Relative absolute error             0 %
Root relative squared error         0 %
Total Number of Instances          500

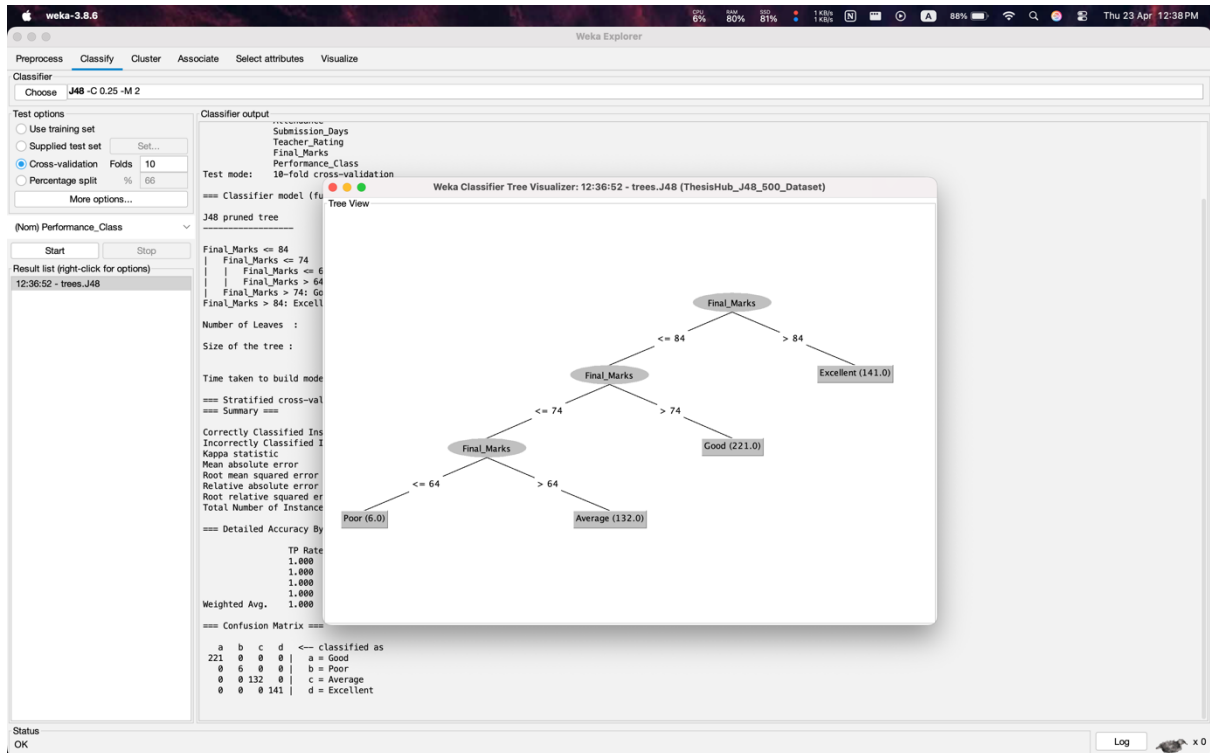
==== Detailed Accuracy By Class ====

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Good
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Poor
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Average
      1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Excellent
Weighted Avg.   1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000

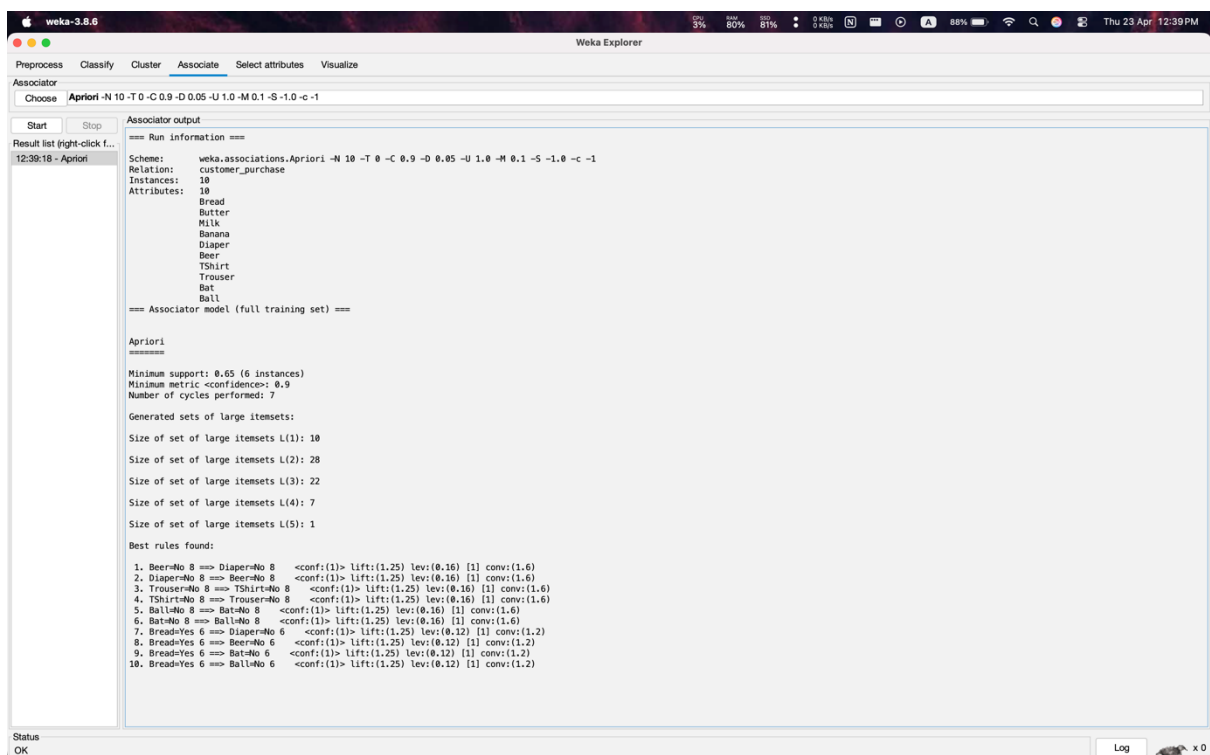
==== Confusion Matrix ====

a b c d  <-- classified as
221 0 0 0  a = Good
0 6 0 0  b = Poor
0 0 132 0  c = Average
0 0 0 141  d = Excellent
```

Status: OK

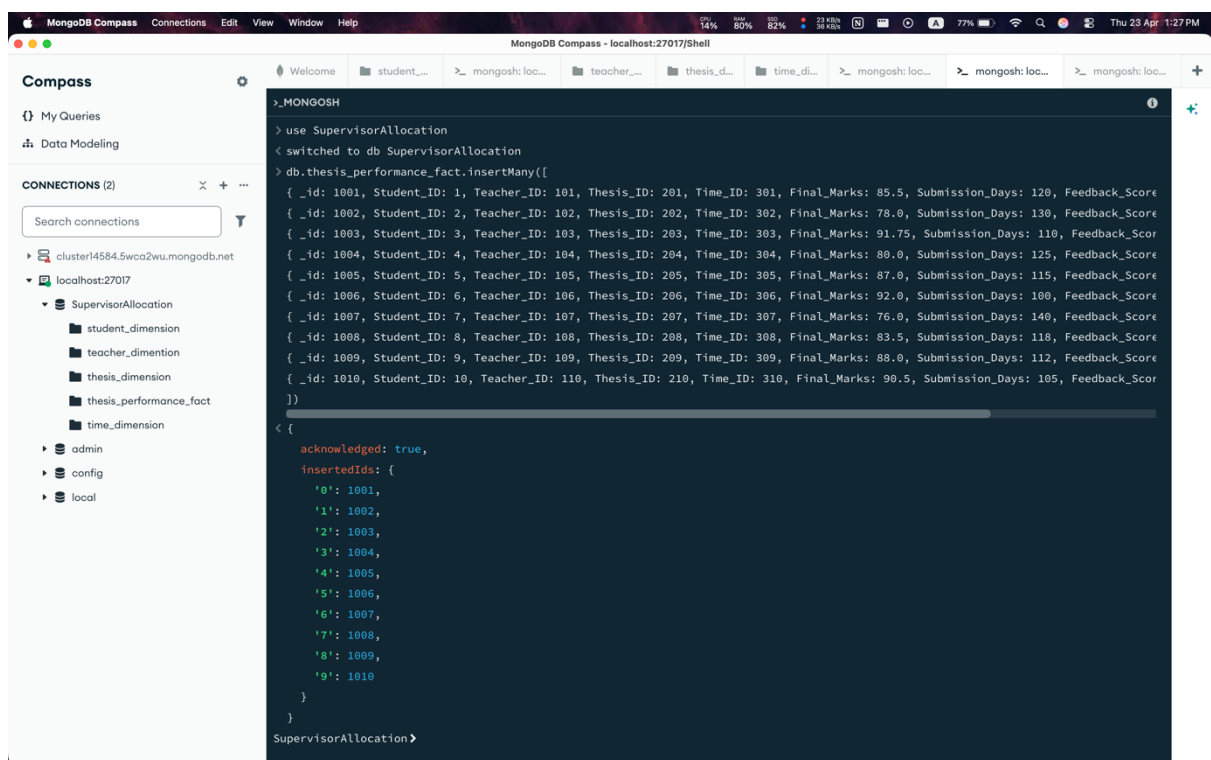


2. Association Rule On my project dataset “Intelligence Automatic Supervisor Allocation System”



3. Design Fact Table and Dimension Table

Fact Table: 1. Creation



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists two connections: 'cluster14584.5wca2wu.mongodb.net' and 'localhost:27017'. Under 'localhost:27017', the 'SupervisorAllocation' database is expanded, showing collections: 'student_dimension', 'teacher_dimension', 'thesis_dimension', 'thesis_performance_fact', and 'time_dimension'. The main panel displays the MongoDB Shell with the following commands and output:

```
> use SupervisorAllocation
switched to db SupervisorAllocation
> db.thesis_performance_fact.insertMany([
  { _id: 1001, Student_ID: 1, Teacher_ID: 101, Thesis_ID: 201, Time_ID: 301, Final_Marks: 85.5, Submission_Days: 120, Feedback_Score: 85.5 },
  { _id: 1002, Student_ID: 2, Teacher_ID: 102, Thesis_ID: 202, Time_ID: 302, Final_Marks: 78.0, Submission_Days: 130, Feedback_Score: 78.0 },
  { _id: 1003, Student_ID: 3, Teacher_ID: 103, Thesis_ID: 203, Time_ID: 303, Final_Marks: 91.75, Submission_Days: 110, Feedback_Score: 91.75 },
  { _id: 1004, Student_ID: 4, Teacher_ID: 104, Thesis_ID: 204, Time_ID: 304, Final_Marks: 80.0, Submission_Days: 125, Feedback_Score: 80.0 },
  { _id: 1005, Student_ID: 5, Teacher_ID: 105, Thesis_ID: 205, Time_ID: 305, Final_Marks: 87.0, Submission_Days: 115, Feedback_Score: 87.0 },
  { _id: 1006, Student_ID: 6, Teacher_ID: 106, Thesis_ID: 206, Time_ID: 306, Final_Marks: 92.0, Submission_Days: 100, Feedback_Score: 92.0 },
  { _id: 1007, Student_ID: 7, Teacher_ID: 107, Thesis_ID: 207, Time_ID: 307, Final_Marks: 76.0, Submission_Days: 140, Feedback_Score: 76.0 },
  { _id: 1008, Student_ID: 8, Teacher_ID: 108, Thesis_ID: 208, Time_ID: 308, Final_Marks: 83.5, Submission_Days: 118, Feedback_Score: 83.5 },
  { _id: 1009, Student_ID: 9, Teacher_ID: 109, Thesis_ID: 209, Time_ID: 309, Final_Marks: 88.0, Submission_Days: 112, Feedback_Score: 88.0 },
  { _id: 1010, Student_ID: 10, Teacher_ID: 110, Thesis_ID: 210, Time_ID: 310, Final_Marks: 90.5, Submission_Days: 105, Feedback_Score: 90.5 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': 1001,
    '1': 1002,
    '2': 1003,
    '3': 1004,
    '4': 1005,
    '5': 1006,
    '6': 1007,
    '7': 1008,
    '8': 1009,
    '9': 1010
  }
}
```

The output shows that the data was successfully inserted into the 'thesis_performance_fact' collection, with 10 documents inserted and their IDs listed in the 'insertedIds' field.

2. Shows data

The image displays two screenshots of the MongoDB Compass interface, showing the 'thesis_performance_fact' collection in the 'SupervisorAllocation' database. The interface includes a sidebar with connections and a main panel for viewing documents.

Top Screenshot: The 'Documents' tab is selected, showing a table with 10 documents. The columns are: `_id` (Int32), `Student_ID` (Int32), `Teacher_ID` (Int32), `Thesis_ID` (Int32), and `Time_ID` (Int).

	<code>_id</code> Int32	<code>Student_ID</code> Int32	<code>Teacher_ID</code> Int32	<code>Thesis_ID</code> Int32	<code>Time_ID</code> Int
1	1001	1	101	201	301
2	1002	2	102	202	302
3	1003	3	103	203	303
4	1004	4	104	204	304
5	1005	5	105	205	305
6	1006	6	106	206	306
7	1007	7	107	207	307
8	1008	8	108	208	308
9	1009	9	109	209	309
10	1010	10	110	210	310

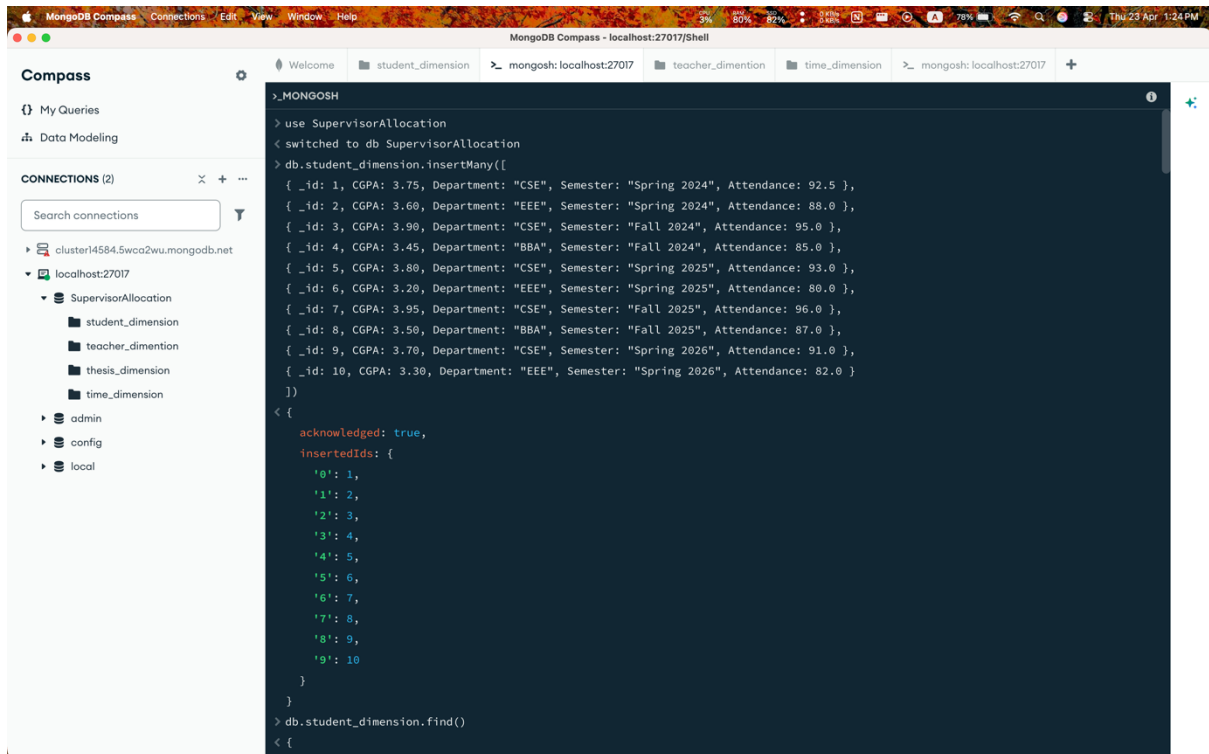
Bottom Screenshot: The 'Documents' tab is selected, showing a table with 10 documents. The columns are: `Final_Marks` (Mixed), `Submission_Days` (Int32), `Feedback_Score` (Double), and `Task_Completion` (Int32).

	<code>Final_Marks</code> Mixed	<code>Submission_Days</code> Int32	<code>Feedback_Score</code> Double	<code>Task_Completion</code> Int32
1	85.5	120	4.6	95
2	78	130	4.2	88
3	91.75	110	4.8	98
4	80	125	4.3	90
5	87	115	4.6	93
6	92	100	4.9	97
7	76	140	4.1	85
8	83.5	118	4.4	89
9	88	112	4.2	92
10	90.5	105	4.7	96

Dimension Table

1. Creation

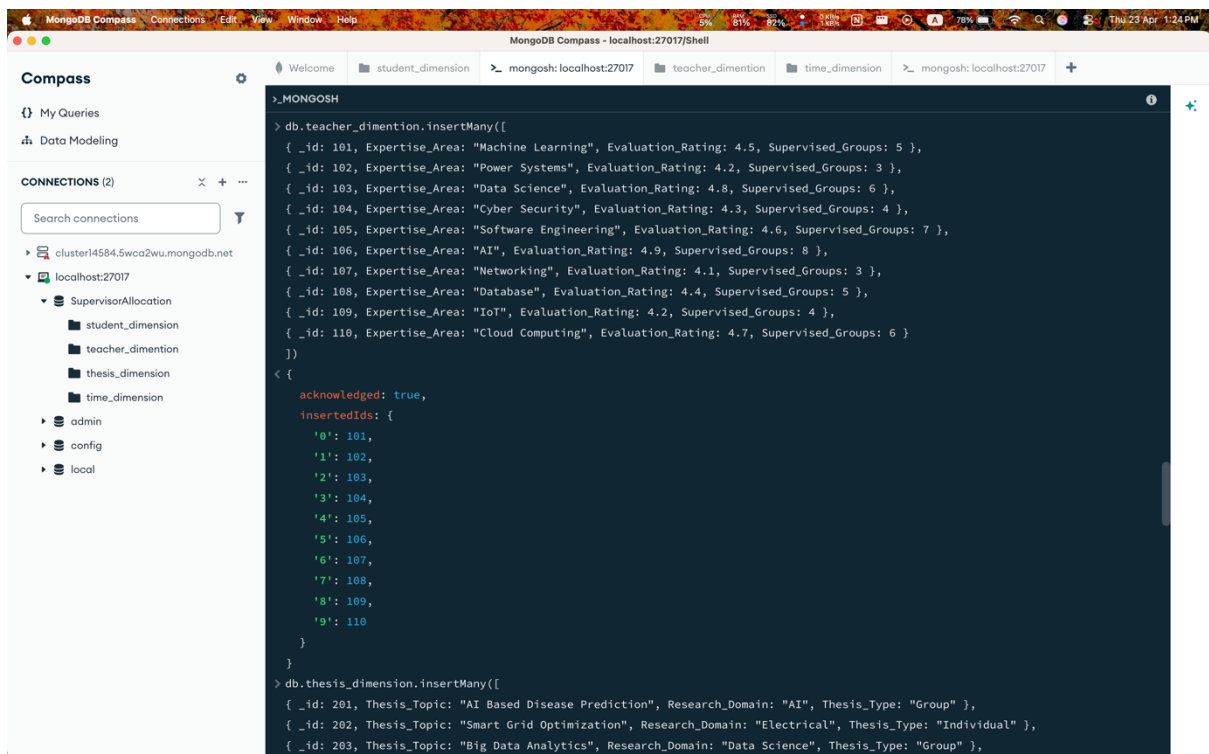
1.1 Student_dimension



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'localhost:27017' connection with a tree view showing databases like 'SupervisorAllocation', 'student_dimension', 'teacher_dimension', 'thesis_dimension', and 'time_dimension'. The main panel displays a MongoDB shell session where the following commands were executed:

```
> use SupervisorAllocation
< switched to db SupervisorAllocation
> db.student_dimension.insertMany([
  { _id: 1, CGPA: 3.75, Department: "CSE", Semester: "Spring 2024", Attendance: 92.5 },
  { _id: 2, CGPA: 3.60, Department: "EEE", Semester: "Spring 2024", Attendance: 88.0 },
  { _id: 3, CGPA: 3.90, Department: "CSE", Semester: "Fall 2024", Attendance: 95.0 },
  { _id: 4, CGPA: 3.45, Department: "BBA", Semester: "Fall 2024", Attendance: 85.0 },
  { _id: 5, CGPA: 3.80, Department: "CSE", Semester: "Spring 2025", Attendance: 93.0 },
  { _id: 6, CGPA: 3.20, Department: "EEE", Semester: "Spring 2025", Attendance: 80.0 },
  { _id: 7, CGPA: 3.95, Department: "CSE", Semester: "Fall 2025", Attendance: 96.0 },
  { _id: 8, CGPA: 3.50, Department: "BBA", Semester: "Fall 2025", Attendance: 87.0 },
  { _id: 9, CGPA: 3.70, Department: "CSE", Semester: "Spring 2026", Attendance: 91.0 },
  { _id: 10, CGPA: 3.30, Department: "EEE", Semester: "Spring 2026", Attendance: 82.0 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5,
    '5': 6,
    '6': 7,
    '7': 8,
    '8': 9,
    '9': 10
  }
}
> db.student_dimension.find()
< {
```

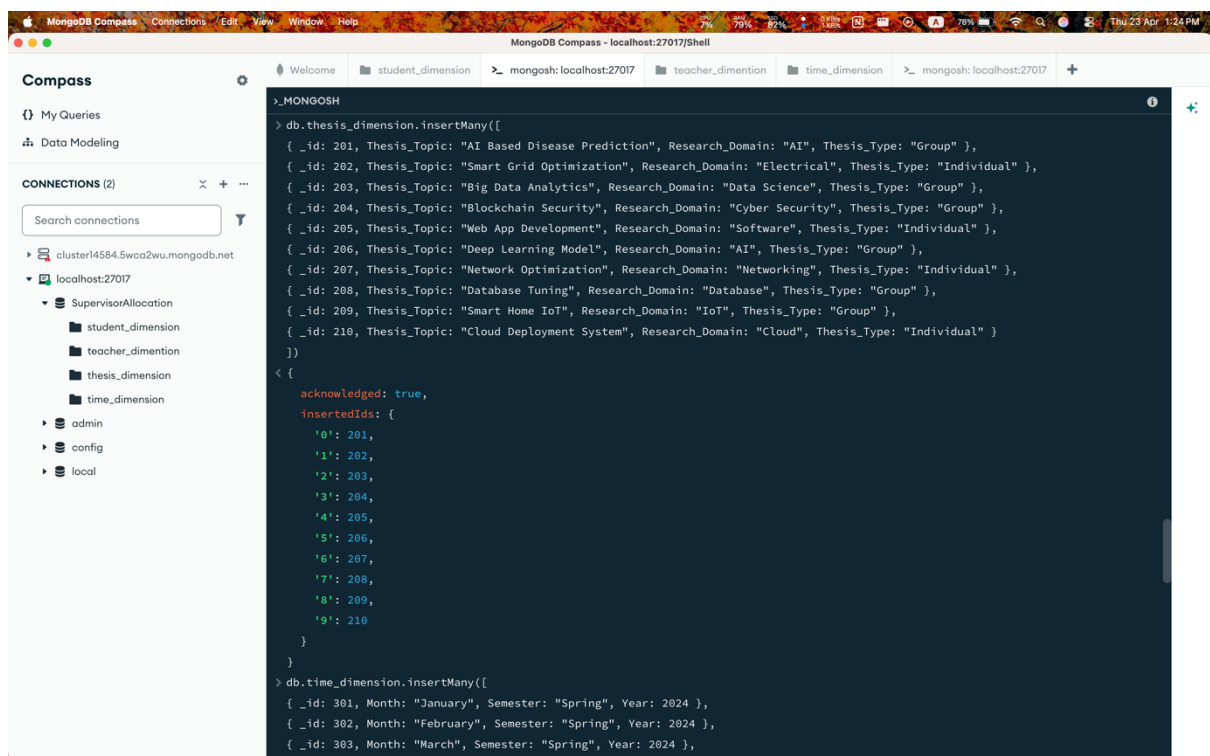
1.2 Teacher_dimension



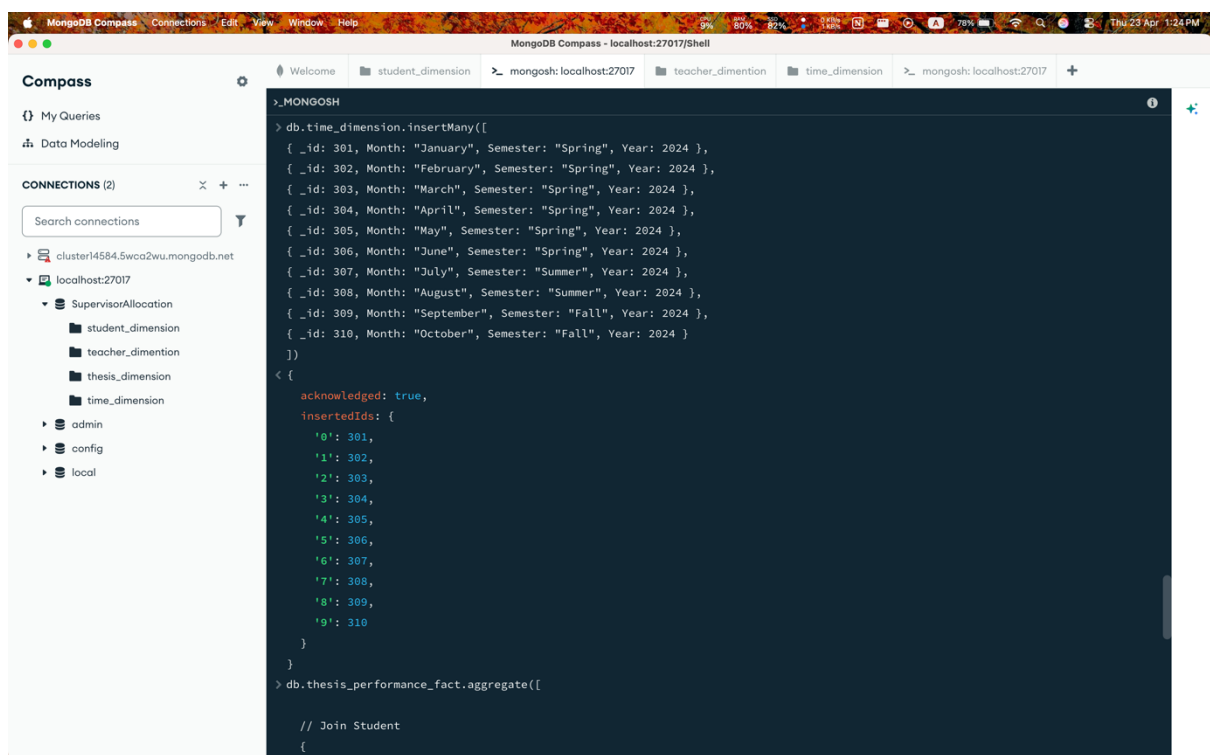
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'localhost:27017' connection with a tree view showing databases like 'SupervisorAllocation', 'student_dimension', 'teacher_dimension', 'thesis_dimension', and 'time_dimension'. The main panel displays a MongoDB shell session where the following commands were executed:

```
> db.teacher_dimension.insertMany([
  { _id: 101, Expertise_Area: "Machine Learning", Evaluation_Rating: 4.5, Supervised_Groups: 5 },
  { _id: 102, Expertise_Area: "Power Systems", Evaluation_Rating: 4.2, Supervised_Groups: 3 },
  { _id: 103, Expertise_Area: "Data Science", Evaluation_Rating: 4.8, Supervised_Groups: 6 },
  { _id: 104, Expertise_Area: "Cyber Security", Evaluation_Rating: 4.3, Supervised_Groups: 4 },
  { _id: 105, Expertise_Area: "Software Engineering", Evaluation_Rating: 4.6, Supervised_Groups: 7 },
  { _id: 106, Expertise_Area: "AI", Evaluation_Rating: 4.9, Supervised_Groups: 8 },
  { _id: 107, Expertise_Area: "Networking", Evaluation_Rating: 4.1, Supervised_Groups: 3 },
  { _id: 108, Expertise_Area: "Database", Evaluation_Rating: 4.4, Supervised_Groups: 5 },
  { _id: 109, Expertise_Area: "IoT", Evaluation_Rating: 4.2, Supervised_Groups: 4 },
  { _id: 110, Expertise_Area: "Cloud Computing", Evaluation_Rating: 4.7, Supervised_Groups: 6 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 101,
    '1': 102,
    '2': 103,
    '3': 104,
    '4': 105,
    '5': 106,
    '6': 107,
    '7': 108,
    '8': 109,
    '9': 110
  }
}
> db.thesis_dimension.insertMany([
  { _id: 201, Thesis_Topic: "AI Based Disease Prediction", Research_Domain: "AI", Thesis_Type: "Group" },
  { _id: 202, Thesis_Topic: "Smart Grid Optimization", Research_Domain: "Electrical", Thesis_Type: "Individual" },
  { _id: 203, Thesis_Topic: "Big Data Analytics", Research_Domain: "Data Science", Thesis_Type: "Group" },
])
```

1.3 Thesis_dimension



1.4 Time_dimension



Show all dimension data:

1.1 Student_dimension Table

The screenshot shows the MongoDB Compass interface with the 'student_dimension' collection selected. The table displays 10 documents with the following columns: _id (Int32), CGPA (Double), Department (String), Semester (String), and Attendance (Double). The data is as follows:

_id	CGPA	Department	Semester	Attendance
1	3.75	"CSE"	"Spring 2024"	92.5
2	3.6	"EEE"	"Spring 2024"	88
3	3.9	"CSE"	"Fall 2024"	95
4	3.45	"BBA"	"Fall 2024"	85
5	3.8	"CSE"	"Spring 2025"	93
6	3.2	"EEE"	"Spring 2025"	80
7	3.95	"CSE"	"Fall 2025"	96
8	3.5	"BBA"	"Fall 2025"	87
9	3.7	"CSE"	"Spring 2026"	91
10	3.3	"EEE"	"Spring 2026"	82

1.2 Teacher_dimension Table

The screenshot shows the MongoDB Compass interface with the 'teacher_dimension' collection selected. The table displays 10 documents with the following columns: _id (Int32), Expertise_Area (String), Evaluation_Rating (Double), and Supervised_Groups (Int32). The data is as follows:

_id	Expertise_Area	Evaluation_Rating	Supervised_Groups
101	"Machine Learning"	4.5	5
102	"Power Systems"	4.2	3
103	"Data Science"	4.8	6
104	"Cyber Security"	4.3	4
105	"Software Engineering"	4.6	7
106	"AI"	4.9	8
107	"Networking"	4.1	3
108	"Database"	4.4	5
109	"IoT"	4.2	4
110	"Cloud Computing"	4.7	6

1.3 Thesis_dimension Table

The screenshot shows the MongoDB Compass interface with the 'thesis_dimension' collection selected. The table contains 10 documents, each with an '_id', 'Thesis_Topic', 'Research_Domain', and 'Thesis_Type'.

_id	Thesis_Topic	Research_Domain	Thesis_Type
201	"AI Based Disease Predict.."	"AI"	"Group"
202	"Smart Grid Optimization"	"Electrical"	"Individual"
203	"Big Data Analytics"	"Data Science"	"Group"
204	"Blockchain Security"	"Cyber Security"	"Group"
205	"Web App Development"	"Software"	"Individual"
206	"Deep Learning Model"	"AI"	"Group"
207	"Network Optimization"	"Networking"	"Individual"
208	"Database Tuning"	"Database"	"Group"
209	"Smart Home IoT"	"IoT"	"Group"
210	"Cloud Deployment System"	"Cloud"	"Individual"

1.4 Time_dimension Table

The screenshot shows the MongoDB Compass interface with the 'time_dimension' collection selected. The table contains 10 documents, each with an '_id', 'Month', 'Semester', and 'Year'.

_id	Month	Semester	Year
301	"January"	"Spring"	2024
302	"February"	"Spring"	2024
303	"March"	"Spring"	2024
304	"April"	"Spring"	2024
305	"May"	"Spring"	2024
306	"June"	"Spring"	2024
307	"July"	"Summer"	2024
308	"August"	"Summer"	2024
309	"September"	"Fall"	2024
310	"October"	"Fall"	2024